

# **Fair & Unbiased Assessment System**

## **Database Design Document**

**V 1.3**

### **Submitted By**

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## REVISION HISTORY

| Date     | Version | Description                                                                                                                                                                                                                                                                             | Approved by   |
|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 15/05/26 | V 1.3   | Finalized the report by updating relationship names, cardinalities and PK identification, references, and AI appendix section. Expanded entity attribute documentation from basic entities to all final entities of the ERD and removed associative entities from the entity list table | Group Members |
| 12/05/26 | V 1.2   | Updated the report according to the modified ERD structure by introducing Program, Batch, Session, and Section entities. Improved normalization, modified Enrollment structure, and updated business rules and assumptions according to university workflow.                            | Ma'am Asiya   |
| 10/05/26 | V 1.1   | Prepared the initial ERD report including introduction, objectives, entity list and attributes. Initial version also included associative entities in the entity table and limited attribute documentation.                                                                             | Self          |

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# CHAPTER 1: PROJECT OVERVIEW

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## 1.1. INTRODUCTION:

The purpose of this milestone is to design and develop a complete Entity Relationship Diagram (ERD/EERD) for the proposed AI-Based Academic Assessment and Report Generation System. This phase focuses on identifying the major entities, defining their attributes, and establishing relationships to represent the complete database structure using standard ERD/EERD notation.

The proposed system is designed to manage academic activities within a university environment, including student enrollment, faculty teaching assignments, course management, assessments, submissions, evaluations, and AI-generated academic reports. The ERD acts as the foundational blueprint of the system by organizing how academic data is stored, connected, and maintained throughout the workflow.

During the design process, special attention was given to real-world university scenarios such as program and batch management, semester and session tracking, section handling, repeat-course enrollment, and preservation of historical academic records. Associative entities such as Enrollment, Teacher Assignment, and Registered Courses were introduced to support accurate academic tracking and maintain proper many-to-many relationships. This milestone provides the structural foundation required for implementing the database system in later phases of the project.

## 1.2. PROBLEM STATEMENT:

Traditional academic management and assessment systems used in many educational institutions mainly focus on storing student records, course information, and examination data while lacking intelligent mechanisms for structured assessment management and automated academic analysis. In most cases, assessment evaluation, feedback generation, and performance analysis are performed manually by instructors, making the process time-consuming, repetitive, and difficult to manage efficiently for large classes.

Additionally, many systems fail to properly maintain academic relationships such as program structures, teaching assignments, enrollment history, batch management, semester progression, and repeat-course scenarios. Poor database design can lead to data redundancy, inconsistency, weak scalability, and difficulties in preserving historical academic records.

The proposed AI-Based Academic Assessment and Report Generation System require a well-structured and normalized database design capable of accurately representing complex university workflows and relationships. Therefore, there is a need to design a comprehensive Entity Relationship Diagram (ERD/EERD) that identifies all major entities, their attributes, associative entities, subtype-supertype structures, and relationships within the system.

The main problems addressed in this milestone include:

- i. Academic management operations
- ii. Assessment and submission workflows
- iii. Faculty teaching assignments
- iv. Student enrollment history
- v. AI-based evaluation processing
- vi. Automated report generation functionalities
- vii. Data consistency and scalability
- viii. Preservation of historical academic records
- ix. Proper representation of real-world university scenarios

### **1.3. PROJECT OBJECTIVES:**

The main objective of this milestone is to design a complete and normalized ERD/EERD for the AI-Based Academic Assessment and Report Generation System that accurately represents academic workflows and relationships within a university environment. The specific objectives of this milestone are as follows:

- 1. To identify all major entities required in the system.
- 2. To define appropriate attributes for each entity.
- 3. To model supertype-subtype relationships for user management.
- 4. To identify associative entities for many-to-many relationships.
- 5. To define relationships and their cardinalities according to university workflows.
- 6. To preserve historical academic records such as enrollments and evaluations.
- 7. To represent departments, programs, batches, semesters, sessions, and sections in a structured manner.
- 8. To support repeat-course enrollment and faculty teaching assignments across programs.
- 9. To support AI-based evaluation and automated report generation.
- 10. To ensure normalization for reducing redundancy and improving consistency.
- 11. To create a professional ERD/EERD using standard Chen notation

### **1.4. GITHUB REPOSITORY LINK:**

Please click the link for the GitHub repository. [ [Link](#) ]

## CHAPTER 2: DETAILED DATABASE DESIGN

### 2.1. ENTITY:

| Sr. No | Entity Name  | Description                                                                                                                        |
|--------|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| 01     | User         | Stores common authentication and profile information for all system users including students, faculty members, and administrators. |
| 02     | Student      | Stores student-specific academic information such as registration numbers, batch, program, and current academic details.           |
| 03     | Faculty      | Stores faculty member information including designation, employee details, and department association.                             |
| 04     | Department   | Stores university department information and manages related programs, faculty members, and courses.                               |
| 05     | Program      | Stores academic program details such as BS AI, BS Cyber Security, and BS IT offered under departments.                             |
| 06     | Batch        | Stores admission batch information of students enrolled in a specific program.                                                     |
| 07     | Semester     | Stores academic semester levels representing student study progression.                                                            |
| 08     | Session      | Stores academic sessions or terms such as Spring 2026 and Fall 2025.                                                               |
| 09     | Section      | Stores section information for a specific program and session.                                                                     |
| 10     | Course       | Stores course-related information including course code, title, and credit hours.                                                  |
| 11     | Assessment   | Stores assessment details such as quizzes, assignments, and deadlines.                                                             |
| 12     | Submission   | Stores student uploaded submissions for assessments.                                                                               |
| 13     | Evaluation   | Stores AI-generated evaluation results, marks, feedback, and performance reports.                                                  |
| 14     | User Phone   | Stores multiple contact numbers associated with system users.                                                                      |
| 15     | Activity Log | Stores user activities and system operation logs.                                                                                  |

### 2.2. DATA DICTIONARY:

#### 2.2.1. User:

| Sr. No | Name                     | Data Type | Constraint          | Description                    |
|--------|--------------------------|-----------|---------------------|--------------------------------|
| 01     | User ID<br>(Primary Key) | Integer   | NOT NULL, PK        | Unique identifier for the user |
| 02     | First Name               | String    | NOT NULL            | First name of the user         |
| 03     | Last Name                | String    | NOT NULL            | Last name of the user          |
| 04     | Email                    | String    | NOT NULL,<br>UNIQUE | Email address of the user      |

|    |                |          |          |                                                     |
|----|----------------|----------|----------|-----------------------------------------------------|
| 05 | Password       | String   | NOT NULL | Encrypted password of the user                      |
| 06 | Role           | String   | NOT NULL | Role of the user (e.g., Admin, Faculty, Student)    |
| 07 | Gender         | String   | NULLABLE | Gender of the user                                  |
| 08 | Account Status | String   | NOT NULL | Status of the user account (e.g., Active, Inactive) |
| 09 | Created At     | DateTime | NOT NULL | Timestamp when the user account was created         |

### 2.2.2. User Phone

| Sr. No | Name                      | Data Type | Constraint   | Description                        |
|--------|---------------------------|-----------|--------------|------------------------------------|
| 01     | phone_ID<br>(Primary Key) | String    | NOT NULL, PK | The unique identifier for the data |
| 02     | Phone_no                  | Integer   | Not Null     | Phone number of the user           |

### 2.2.3. Student

| Sr. No | Name                        | Data Type | Constraint       | Description                                               |
|--------|-----------------------------|-----------|------------------|-----------------------------------------------------------|
| 01     | Student_ID<br>(Primary Key) | Integer   | NOT NULL, PK     | Unique identifier for the student                         |
| 02     | registration_no             | String    | NOT NULL, UNIQUE | Official registration number of the student               |
| 03     | current_semester            | Integer   | NOT NULL         | Current semester of the student                           |
| 04     | student_status              | String    | NOT NULL         | Current status of the student (e.g., Enrolled, Graduated) |

### 2.2.4. Faculty

| Sr. No | Name                        | Data Type | Constraint   | Description                                |
|--------|-----------------------------|-----------|--------------|--------------------------------------------|
| 01     | Teacher_ID<br>(Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the faculty member   |
| 02     | Designation                 | String    | NOT NULL     | Academic designation of the faculty member |

### 2.2.5. Program

| Sr. No | Name                        | Data Type | Constraint       | Description                                     |
|--------|-----------------------------|-----------|------------------|-------------------------------------------------|
| 01     | program_ID<br>(Primary Key) | Integer   | NOT NULL, PK     | Unique identifier for the academic program      |
| 02     | program_name                | String    | NOT NULL         | Name of the academic program                    |
| 03     | program_code                | String    | NOT NULL, UNIQUE | Short code for the program                      |
| 04     | degree_level                | String    | NOT NULL         | Level of degree (e.g., Undergraduate, Graduate) |
| 05     | total_semester              | Integer   | NOT NULL         | Total number of semesters in the program        |

### 2.2.6. Activity Log

| Sr. No | Name                 | Data Type | Constraint   | Description                                  |
|--------|----------------------|-----------|--------------|----------------------------------------------|
| 01     | Log_id (Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the activity log entry |
| 02     | Activity_description | Text      | NOT NULL     | Detailed description of the activity         |
| 03     | Time                 | DateTime  | NOT NULL     | Timestamp when the activity occurred         |

### 2.2.7. Department

| Sr. No | Name                           | Data Type | Constraint       | Description                            |
|--------|--------------------------------|-----------|------------------|----------------------------------------|
| 01     | Department_ID<br>(Primary Key) | Integer   | NOT NULL, PK     | Unique identifier for the department   |
| 02     | Department Name                | String    | NOT NULL         | Full name of the department            |
| 03     | Department_Code                | String    | NOT NULL, UNIQUE | Short code representing the department |

### 2.2.8. Batch

| Sr. No | Name                   | Data Type | Constraint   | Description                            |
|--------|------------------------|-----------|--------------|----------------------------------------|
| 01     | Batch_ID (Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the batch        |
| 02     | Batch_name             | String    | NOT NULL     | Name of the batch                      |
| 03     | Expected_Graduation    | Date      | NULLABLE     | Expected graduation date for the batch |

|    |              |         |          |                             |
|----|--------------|---------|----------|-----------------------------|
| 04 | Batch_status | String  | NOT NULL | Current status of the batch |
| 05 | start_year   | Integer | NOT NULL | The year the batch started  |

### 2.2.9. Section

| Sr. No | Name                        | Data Type | Constraint   | Description                                       |
|--------|-----------------------------|-----------|--------------|---------------------------------------------------|
| 01     | Section ID<br>(Primary Key) | Integer   | NOT NULL, PK | ID or identifier for the section                  |
| 02     | capacity                    | Integer   | NOT NULL     | Maximum number of students allowed in the section |
| 03     | Section Name                | String    | Not Null     | Name or identifier for the section                |

### 2.2.10. Session

| Sr. No | Name                        | Data Type | Constraint   | Description                       |
|--------|-----------------------------|-----------|--------------|-----------------------------------|
| 01     | session_ID<br>(Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the session |
| 02     | session_name                | String    | NOT NULL     | Name of the session               |
| 03     | Academic_year               | String    | NOT NULL     | Academic year of the session      |
| 04     | start_date                  | Date      | NOT NULL     | Start date of the session         |
| 05     | end_date                    | Date      | NOT NULL     | End date of the session           |

### 2.2.11. Teacher Assignment

| Sr. No | Name                           | Data Type | Constraint   | Description                                  |
|--------|--------------------------------|-----------|--------------|----------------------------------------------|
| 01     | Assignment_ID<br>(Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the teacher assignment |
| 02     | Assigned Date                  | Date      | NOT NULL     | Date when the assignment was made            |

### 2.2.12. Course

| Sr. No | Name                       | Data Type | Constraint       | Description                      |
|--------|----------------------------|-----------|------------------|----------------------------------|
| 01     | Course_ID<br>(Primary Key) | Integer   | NOT NULL, PK     | Unique identifier for the course |
| 02     | Course_Code                | String    | NOT NULL, UNIQUE | Official code for the course     |

|    |              |         |          |                                       |
|----|--------------|---------|----------|---------------------------------------|
| 03 | Course_Title | String  | NOT NULL | Full title of the course              |
| 04 | Credit_Hours | Integer | NOT NULL | Number of credit hours for the course |

### 2.2.13. Enrollment

| Sr. No | Name                           | Data Type | Constraint   | Description                                        |
|--------|--------------------------------|-----------|--------------|----------------------------------------------------|
| 01     | Enrollment_ID<br>(Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the enrollment record        |
| 02     | is_repeat                      | Boolean   | NOT NULL     | Indicates if the course is being repeated          |
| 03     | Enrollment_Status              | String    | NOT NULL     | Status of the enrollment (e.g., Pending, Approved) |
| 04     | Enrollment_Start_Date          | Date      | NOT NULL     | Date when enrollment started                       |
| 05     | Enrollment_End_Date            | Date      | NULLABLE     | Date when enrollment ended                         |

### 2.2.14. Semester

| Sr. No | Name                         | Data Type | Constraint   | Description                                |
|--------|------------------------------|-----------|--------------|--------------------------------------------|
| 01     | Semester_ID<br>(Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the semester         |
| 02     | Semester_no                  | Integer   | NOT NULL     | The number of the semester in the sequence |
| 03     | Academic_year                | String    | NOT NULL     | The academic year of the semester          |
| 04     | start_date                   | Date      | NOT NULL     | Start date of the semester                 |
| 05     | end_date                     | Date      | NOT NULL     | End date of the semester                   |

### 2.2.15. Assessment

| Sr. No | Name                           | Data Type | Constraint   | Description                            |
|--------|--------------------------------|-----------|--------------|----------------------------------------|
| 01     | Assessment_ID<br>(Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the assessment   |
| 02     | Title                          | String    | NOT NULL     | Title of the assessment                |
| 03     | Description                    | Text      | NULLABLE     | Detailed description of the assessment |

|    |                           |          |          |                                                   |
|----|---------------------------|----------|----------|---------------------------------------------------|
| 04 | Assessment_Type           | String   | NOT NULL | Type of assessment (e.g., Quiz, Assignment, Exam) |
| 05 | Total Marks               | Integer  | NOT NULL | Maximum marks possible for the assessment         |
| 06 | Created_At                | DateTime | NOT NULL | Timestamp when the assessment was created         |
| 07 | Deadline                  | DateTime | NOT NULL | Submission deadline for the assessment            |
| 08 | class_report_path         | String   | NULLABLE | File path to the generated class report           |
| 09 | class_report_generated_at | DateTime | NULLABLE | Timestamp when the class report was generated     |
| 10 | class_report_statuses     | String   | NULLABLE | Current status of the class report generation     |

#### 2.2.16. Registered Course

| Sr. No | Name                               | Data Type | Constraint   | Description                                        |
|--------|------------------------------------|-----------|--------------|----------------------------------------------------|
| 01     | Registered_Course_ID (Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the registered course record |

#### 2.2.17. Evaluation

| Sr. No | Name                        | Data Type | Constraint   | Description                                            |
|--------|-----------------------------|-----------|--------------|--------------------------------------------------------|
| 01     | Evaluation_ID (Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the evaluation                   |
| 02     | Obtained_Marks              | Decimal   | NOT NULL     | Marks obtained by the student                          |
| 03     | Percentage                  | Decimal   | NOT NULL     | Percentage score calculated from marks                 |
| 04     | Evaluated_At                | DateTime  | NOT NULL     | Timestamp when the evaluation was performed            |
| 05     | Evaluation_Status           | String    | NOT NULL     | Status of the evaluation (e.g., Graded, Re-evaluating) |
| 06     | student_report_path         | String    | NULLABLE     | File path to the student's individual report           |

|    |                             |          |          |                                                 |
|----|-----------------------------|----------|----------|-------------------------------------------------|
| 07 | student_report_generated_at | DateTime | NULLABLE | Timestamp when the student report was generated |
| 08 | student_report_status       | String   | NULLABLE | Status of the student report generation         |

### 2.2.18. Submission

| Sr. No | Name                           | Data Type | Constraint   | Description                                      |
|--------|--------------------------------|-----------|--------------|--------------------------------------------------|
| 01     | Submission_ID<br>(Primary Key) | Integer   | NOT NULL, PK | Unique identifier for the submission             |
| 02     | File_Name                      | String    | NOT NULL     | Name of the submitted file                       |
| 03     | File_Path                      | String    | NOT NULL     | Storage path of the submitted file               |
| 04     | Upload_Time                    | DateTime  | NOT NULL     | Timestamp when the file was uploaded             |
| 05     | Submission Status              | String    | NOT NULL     | Status of the submission (e.g., Submitted, Late) |

## 2.3. BUSINESS RULES:

- Each user in the system must have a unique user account for authentication and system access.
- A user can exist either as a student or a faculty member within the system.
- A user can have multiple contact numbers stored in the system.
- Each student must belong to one specific program throughout the degree duration.
- Each student must belong to one admission batch of a program.
- A department can offer multiple academic programs, but each program belongs to only one department.
- A department can offer multiple courses, but each course belongs to only one department.
- A program can contain multiple sections, but each section belongs to one specific program.
- Each section is conducted within one academic session, while a session can contain multiple sections.
- A faculty member belongs to one department but can teach multiple courses and sections.
- A faculty member can have multiple teaching assignments in different semesters and sections.
- Each teaching assignment is associated with one faculty member, one course, one section, and one semester.

- A course can be taught in multiple sections and semesters through different teaching assignments.
- A student can have multiple enrollment records during different semesters and sessions.
- Each enrollment record belongs to one student and stores the student's semester, section, and session information.
- A single enrollment can contain multiple registered courses through Course Enrollment records.
- Each Course Enrollment record links one enrollment with one teaching assignment.
- A student can repeat a failed course in another semester or section through a new enrollment record.

## 2.4. ENTITY RELATIONSHIP DIAGRAM:

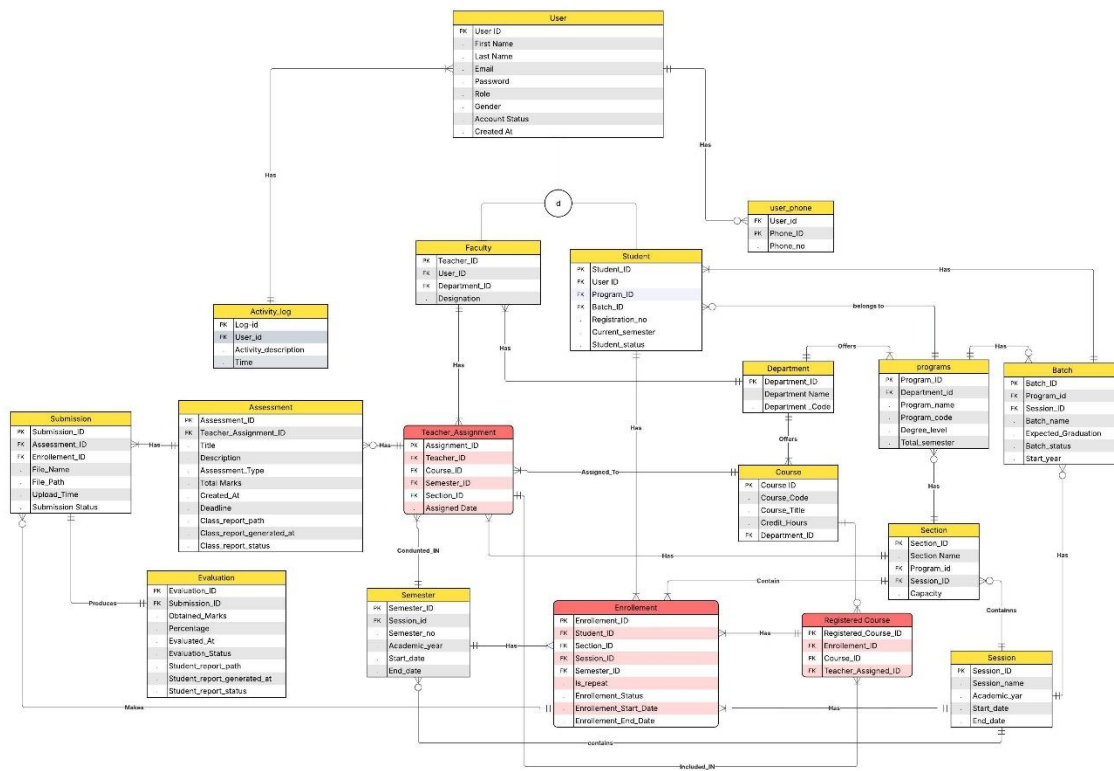


Figure 1: Enhanced Entity Relationship Diagram

### References

- [1] Lucid Software Inc., “What is an Entity Relationship Diagram (ERD)?,” *Lucidchart*, 2024. [Online]. Available: <https://www.lucidchart.com/pages/er-diagrams>. [Accessed: 18-May-2026].
- [2] Oracle Corporation, “Database Design Basics,” *Oracle Documentation*, 2024. [Online]. Available: <https://docs.oracle.com/en/database/>. [Accessed: 18-May-2026].
- [3] Creately, “Entity Relationship Diagram Symbols and Notation,” *Creately*, 2024. [Online]. Available: <https://creately.com/guides/er-diagram-symbols/>. [Accessed: 18-May-2026].

## Chapter 4 APPENDIX: AI USAGE AND PROMPTS

### 4.1 PURPOSE OF AI USAGE

Artificial Intelligence (AI) tools were used during the design and documentation phase of the Entity Relationship Diagram (ERD/EERD) milestone for the proposed AI-Based Academic Assessment and Report Generation System. The AI tool was mainly used for ERD consultation, relationship validation, normalization guidance, business rule formulation, and report writing assistance.

The final ERD structure, assumptions, and academic workflow decisions were finalized according to the project requirements and university scenarios.

### 4.2 AI TOOL USED

| Tool Name        | Purpose                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------|
| ChatGPT (OpenAI) | ERD consultation, relationship validation, normalization guidance, and report writing assistance |

### 4.3 PROMPTS USED DURING DEVELOPMENT

| Sr. No | Prompt Used                                                                                          | Purpose                                                 |
|--------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1      | "Now tell me the final entities and attributes."                                                     | To finalize the entities and attributes of the ERD      |
| 2      | "The enrollment is associative and how many entities links to it?"                                   | To understand associative entity relationships          |
| 3      | "Now from this enhancement give me other relation also carefully understanding the system."          | To refine relationship structures                       |
| 4      | "What is the cardinality of the relation with program and session?"                                  | To validate cardinalities and participation constraints |
| 5      | "Now analyze this ERD and tell me. Analyze in all scenarios and assumptions."                        | To validate the final ERD design                        |
| 6      | "Now you have my ERD but some relation names missing in this so tell me every single relation name." | To finalize relationship naming                         |
| 7      | "Write 15–25 strong rules according to your system."                                                 | Generating business rules                               |
| 8      | "Add assumptions very carefully because this will defend my ERD."                                    | To document system assumptions                          |
| 9      | "Explain supertype/subtype according to my ERD."                                                     | To validate inheritance modeling                        |
| 10     | "Make the proper flow of this report headings."                                                      | To structure the milestone report                       |
| 11     | "Mention all the entities in table form."                                                            | To prepare report documentation                         |
| 12     | "Mention all the attributes in table form for each entity."                                          | To document entity attributes professionally            |

## **4.4 AI USAGE SUMMARY**

The AI tool assisted in identifying and refining entities, designing associative entities, validating cardinalities and relationships, creating business rules and assumptions, improving normalization decisions, and structuring the ERD report professionally.

## **4.5 LINK OF PROMPTS**

You can access the full chat history and prompt interactions at the following link: [[CHAT LINK HERE](#)]